

The Prime Side of a Weil Witness from the Icosian Triad: the Gap, Localized

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Read first. The witness here is **not** a witness for the classical Riemann zeta function. It is a Weil witness for a *specific cuspidal level-31 L -function* attached to the object's Brandt face; classical ζ sits in a separate (Eisenstein) face. Its universal positivity quantifier is therefore **GRH for that cuspidal L -function, not classical RH**, and its relevance to RH is **structural, not a reduction**. We supply the parameter-free *prime side*; the archimedean side is the standard completion of the cuspidal L -function. No claim about RH is made.

Abstract

The companion papers establish the icosian object exactly: the maximal icosian order $\mathcal{I}$ on the 600-cell over $K = \mathbb{Q}(\sqrt{5})$, its triad $(\mathcal{I}, C_\varphi, N_H)$ with Galois symmetry σ , and its Eisenstein theta L -function $L(\Theta_{\mathcal{I}}, s) = \zeta_K(s)\zeta_K(s-1)$. Here we show that the same object also supplies the prime-side ingredient of a Weil positivity witness $W(h)$, parameter-free, and we locate precisely the single step it does not close. The prime residual is assembled entirely from the geometric eigenvalues of the self-adjoint icosian Brandt operator at level $\mathfrak{p}_{31} = (5\varphi - 2)$ (the object's *cuspidal* face), with no fitting and no zero-data; the archimedean term is the cuspidal L -function's completion (gamma factor and conductor), fixed by its functional equation. The implementation is validated on ζ (where the zeros are known) and, on a finite Gaussian test family, returns nonnegative values of the truncated computed functional $W_{\leq 150}(h)$ (prime side cut at $N(\mathfrak{q}) \leq 150$). **We make no claim about the Riemann Hypothesis.** Crucially: the witness lives on the *cuspidal* face, so its positivity quantifier is GRH for that level-31 L -function, *not* classical RH (classical ζ sits in the separate Eisenstein face). The verified tests are moreover archimedean-dominated, so they are consistent with, but not a sharp test of, that quantifier. The result is a *localization*, not a reduction of classical RH.

1 The two arithmetic faces of one object

The icosian order $\mathcal{I}$ carries two distinct arithmetic objects, and keeping them separate is the whole point of this note.

Eisenstein face (theta). The theta series of $\mathcal{I}$ has $L(\Theta_{\mathcal{I}}, s) = \zeta_K(s)\zeta_K(s-1)$ exactly [1]; via $\zeta_K = \zeta \cdot L(s, \chi_5)$ the Riemann ζ appears as one of its four factors. This face *contains* classical ζ . Its parameters are $a_{\mathfrak{q}} = N(\mathfrak{q}) + 1$, which violate the Ramanujan bound $|a_{\mathfrak{q}}| \leq 2\sqrt{N(\mathfrak{q})}$, so it is *not admissible as the unitary cuspidal Satake input* used in the witness below. (It still has an explicit formula of its own; we make no use of one here.)

Cuspidal face (Brandt). The same order carries a self-adjoint Brandt/Hecke operator $B(\mathfrak{q})$ at level $\mathfrak{p}_{31} = (5\varphi - 2)$, whose eigenvalues $a_{\mathfrak{q}}$ are read directly from the icosian Brandt matrices—no point-counting, no fitting. (In the external route-B computation [2] these eigenvalues were further checked out-of-sample against point counts; that comparison lives in that bundle, not this one.) The computed $a_{\mathfrak{q}}$ obey the Ramanujan bound in the tested range; global Ramanujan for the associated form follows from its Hilbert-modular / Jacquet–Langlands identification, not from the finite Brandt matrices alone. This cuspidal face is where the witness lives.

2 The Weil witness, and that its prime side is built from the object

Fix the cuspidal level-31 L -function of the previous paragraph, with completed form $\Lambda(s) = N^{s/2} L_{\infty}(s) L(s)$ (gamma factor L_{∞} and conductor $N = 31 \cdot 25 = 775$). For a test function h even and of *positive type* (the Fourier transform $\hat{h} \geq 0$), Weil’s explicit formula reads, schematically,

$$W(h) = \underbrace{\text{ARCH}(h)}_{\text{from } L_{\infty}, \text{ conductor}} - \underbrace{\text{PRIME}(h)}_{\text{from the } a_{\mathfrak{q}}} = \sum_{\rho} \hat{h}(\rho),$$

the sum running over the nontrivial zeros ρ of $L(s)$, *with no assumption on $\text{Re } \rho$* [3]. Writing $\rho = \frac{1}{2} + i\gamma$ with $\gamma \in \mathbb{R}$ is precisely GRH for this L , so we do not presuppose it. The witness W is unconditional; Weil’s criterion is the theorem

$$\text{GRH for this cuspidal } L \iff W(h) \geq 0 \text{ for every } h \text{ of positive type.}$$

Proposition 1 (the prime side is parameter-free from the object). *In the computation [2], $\text{PRIME}(h)$ is assembled entirely from the geometric Brandt eigenvalues $a_{\mathfrak{q}}$ of $\mathcal{I}$ ’s cuspidal face, over the cached prime ideals $N(\mathfrak{q}) \leq 150$; the archimedean term $\text{ARCH}(h)$ is the cuspidal L -function’s completion (gamma type and conductor 775), fixed by its functional equation. No constant is fitted, and no zero-data of the target enters the construction.*

So the *prime side* of the witness is derived from the icosian triad—it is not placed beside it. The archimedean side is the automorphic completion of the chosen cuspidal form, not itself read off the geometry; we state this division of inputs explicitly rather than claiming the whole witness is geometric.

3 What is computed, and the control

The construction (`weil_wall.py`, sourcing `geometric_aP.py`) was run as follows. All tests are Gaussian, $h(r) = e^{-r^2/2\sigma_h^2}$ with $\sigma_h \in \{3, 4, 5, 6\}$.

Computed Result 2 (validation on ζ). Run on the Riemann ζ , where the zeros are known, the same implementation reproduces $\sum_{\gamma} h(\gamma)$ to relative error $\sim 10^{-2}$ down to 10^{-6} across $\sigma_h = 3, \dots, 6$. This validates the normalizations and pole/gamma bookkeeping; no constants are fitted. (The ζ run uses ζ ’s own prime term, pole, and gamma factor; it shares the framework, not the data.)

Computed Result 3 (positivity on the tested family). For the cuspidal level-31 L -function, the truncated computed functional $W_{\leq 150}(h) \geq 0$ (prime side cut at $N(\mathfrak{q}) \leq 150$, no tail bound) on every test in the family ($\sigma_h = 3, 4, 5, 6$; $W_{\leq 150} = +1.4047, +3.6961, +6.3931, +9.4125$). This is the implementation’s value on a finite family, not the full Weil functional $W(h)$.

Computed Result 4 (control). Feeding the *Eisenstein* bookkeeping $a_q = N(q) + 1$ (the theta-face parameters) into the same residual yields 32 Ramanujan violations: these coefficients are not the admissible cuspidal Satake input the residual requires. The control thus has teeth—the Eisenstein input is not admissible for this cuspidal Satake residual—without implying the Eisenstein face is disqualified from all explicit-formula positivity.

4 The gap, named precisely

Two statements fix the honest scope.

Remark 5 (the quantifier is GRH for the cuspidal L , not classical RH). $W(h) \geq 0$ for *all* h of positive type is, by Weil’s criterion, GRH for this cuspidal level-31 L -function—not classical RH. Classical ζ lives in the Eisenstein face (§1), which is not the witness face. The verified positivity is on a finite Gaussian family; the universal quantifier is the wall and is GRH-equivalent, not closable without the (out-of-reach) operator/positivity proof.

Remark 6 (the tests are archimedean-dominated). Against our own result: in the tested family the prime residual is tiny relative to the archimedean term (ratio $\sim 10^{-4}$ down to 10^{-16} as σ_h grows), and the prime term is truncated to the cached range $N(\mathfrak{q}) \leq 150$ with no tail bound. **The numerical tests validate normalization and implementation; they are not evidence for the universal positivity quantifier.** $W(h) \geq 0$ here is consistent with GRH for this L -function but is not a sharp test of it; we supply no sharp test family.

Remark 7 (what a sharp test would require—future work). A genuine test of the quantifier needs: (i) *adversarial* positive-type test functions h for which $\text{PRIME}(h)$ is comparable to $\text{ARCH}(h)$ (so the prime side can actually threaten positivity), and (ii) a certified *tail bound* on the truncated prime residual beyond $N(\mathfrak{q}) \leq 150$. Neither is supplied here; both are the next required steps before any positivity statement carries weight.

5 Conclusion

The icosian triad is exact at the object level, and it supplies—parameter-free, from its cuspidal Brandt face—the prime side of a Weil positivity witness $W(h)$ for a specific cuspidal level-31 L -function, with the archimedean side taken from that form’s completion. The implementation is validated on ζ and gives $W_{\leq 150}(h) \geq 0$ (the truncated functional, $N(\mathfrak{q}) \leq 150$) on every test in a finite Gaussian family. The step not closed is the universal positivity quantifier, which is **GRH for that cuspidal L -function**; and even the verified positivity sits in an archimedean-dominated, finitely-truncated regime that is not a sharp test of it.

The localization is therefore *not* a reduction of classical RH. After choosing the cuspidal Brandt face, the remaining wall is GRH for that one L -function—one named inequality on one parameter-free prime residual. Classical ζ remains in the separate Eisenstein face, which is not positivity-valid for this residual. We make no claim about the Riemann Hypothesis. One object, two arithmetic faces—Eisenstein (theta, containing ζ) and cuspidal (Brandt, carrying the witness)—and the wall is the positivity quantifier on the second.

References

- [1] VFD programme, *The Icosian Triad on V_{600}* (icosian-triad-v600 release, v1.1): $L(\Theta_{\mathcal{I}}, s) = \zeta_K(s)\zeta_K(s-1)$ exactly, verified in the maximal order; states it does not address RH/GRH.

- [2] VFD programme, route-B icosian Brandt computation (external icosian_brandt_cuspidal_geometry/route_b source; self-contained copies `weil_wall.py`, `geometric_aP.py` ship in this bundle's `repro/src`): geometric Hecke eigenvalues a_q of the self-adjoint icosian Brandt operator at level $\mathfrak{p}_{31}$, cache $N(\mathfrak{q}) \leq 150$, out-of-sample checked against point counts; the witness run uses conductor 775 and is archimedean-dominated.
- [3] A. Weil, *Sur les “formules explicites” de la théorie des nombres premiers*.